

Software Testing, Quality Assurance, and Continuous Integration

COMP-7431: Advanced Software Engineering | Week 10

CampusConnect: prove the multimodal MVP before freezing a release candidate

Test exact behavior. Evaluate AI behavior. Gate the release with evidence.

Monday Morning: Week 9 Built It—Does It Deserve a Freeze?

Week 9 produced one traceable CampusConnect journey: grounded RAG, visible sources, optional voice, prerecorded video, and text fallback.

WHAT THE TEAM CAN DEMONSTRATE

“A student can ask how to reset a campus password, receive an answer with a source, hear the same answer, and continue if media is unavailable.”

WEEK 10 QUALITY QUESTION

Can the team show—repeatedly and with evidence—that exact software behavior works, AI answers remain grounded, failures preserve control, and severe defects block release?

A successful demo is evidence of possibility. A release standard is evidence of dependable behavior under defined conditions.

One MVP Contains Three Different Quality Problems

EXACT SOFTWARE BEHAVIOR

Some outcomes should be identical every time: required fields validate, the source link renders, captions remain available, and a failed voice request preserves text.

Example: status=voice_unavailable must still return answer_text and citation_url.

VARIABLE AI BEHAVIOR

The wording may change while quality remains acceptable. The team scores grounding, retrieval relevance, citation support, and appropriate abstention.

Example: “Change my password” and “reset my login” may produce different sentences but should use the same approved policy.

RELEASE DECISION

The architect combines test results, evaluation scores, accessibility evidence, latency, and defect severity into one go/no-go standard.

Example: a missing comma may wait; an invented source or inaccessible essential instruction may block the freeze.

The mistake is not variation itself. The mistake is testing variable AI output as exact text—or treating exact software failures as subjective.

Week 10 Makes Quality an Architectural Decision

QUALITY DECISIONS

- Separate deterministic tests from probabilistic evaluation
- Build a representative CampusConnect evaluation set
- Define expected outcomes, thresholds, and fallbacks before running tests

Example: password-reset paraphrases may vary in wording, but every acceptable answer must use approved evidence and show the correct source.

RELEASE EVIDENCE

- Use GitHub Actions for repeatable software checks
- Use Lighthouse plus manual accessibility review
- Prioritize defects and define what blocks the release candidate

Example: CI may pass the code while a manual muted-video check reveals that the transcript omits the only next step.

The architect does not merely collect scores. The architect decides what the evidence means and what the team must do next.

Testing, AI Evaluation, and QA + CI Do Different Jobs

SOFTWARE TESTING

Check exact behavior against a known expectation.

EXAMPLE

Assert that a voice timeout retains the typed question, answer text, and source link.

AI EVALUATION

Score variable output against defined criteria and representative cases.

EXAMPLE

Judge whether three password-reset paraphrases retrieve the correct passage and stay grounded.

QUALITY ASSURANCE + CI

Define the process and repeat selected checks on repository changes.

EXAMPLE

A pull request cannot become the release candidate while required software checks fail or a release-blocking defect remains open.

A green CI run means the configured checks passed on that commit. It does not prove that the team selected every important quality question.

Move from Quality Claim to Release Gate

1 DEFINE

Name the behavior and risk

Example: unsupported questions must not invent campus policy.

2 MEASURE

Choose an exact test or scored evaluation

Example: assert the fallback field; score grounding and citation support.

3 TRIAGE

Record severity, owner, evidence, and decision

Example: invented citation = release blocker; cosmetic spacing = defer.

4 GATE

Prevent freeze until required evidence passes

Example: GitHub checks are green and no blocker remains open.

The team should be able to trace every release gate back to a user, system, accessibility, or operational risk.

The AI Periodic Table Places Week 10 Across the Quality Path

RETRIEVAL + RAG → COLUMN III

Test whether the system finds the right campus passage before judging the final prose.

A weak answer may begin with irrelevant retrieval—not with the model itself.

EVALUATION → COLUMN IV

Measure output quality with defined cases, criteria, scoring, and documented limits.

CampusConnect scores relevance, grounding, citation support, and abstention.

GUARDRAILS → COLUMN VI

Verify that unsupported or sensitive questions reach a bounded response and useful next step.

A guardrail must work in ordinary use before Week 11 tries to defeat it.

COLUMNS VII + XI

Test contracts, latency, fallbacks, accessibility, and the complete user journey.

GitHub preserves the evidence; GitHub Actions repeats the deterministic checks.

The modern architect works chiefly at System Integration and Applied AI Systems—while drawing quality evidence from earlier AI stages.

Exact Contracts Deserve Exact Tests

DETERMINISTIC TEST STATEMENT

“Given the same controlled input and system state, the software should produce the same observable result every time.”

- Required response fields exist: answer_text, citation_url, status, fallback
- Unsupported input returns the defined abstention state
- A voice timeout preserves text and source
- The transcript is present beside the video
- The Play control has an accessible name
- No secret value appears in browser code or committed files

Do not use a subjective rubric when an assertion can settle the question. Exact contracts deserve exact evidence.

Write Exact Criteria at the Boundary the User Depends On

BEHAVIOR CRITERION

State the observable result—not the private implementation steps.

Example: A missing question produces a helpful validation message and never sends a retrieval request.

CONTRACT CRITERION

State the required fields, state transition, and preserved data.

Example: voice_unavailable retains question, answer_text, citation_url, and Continue in text.

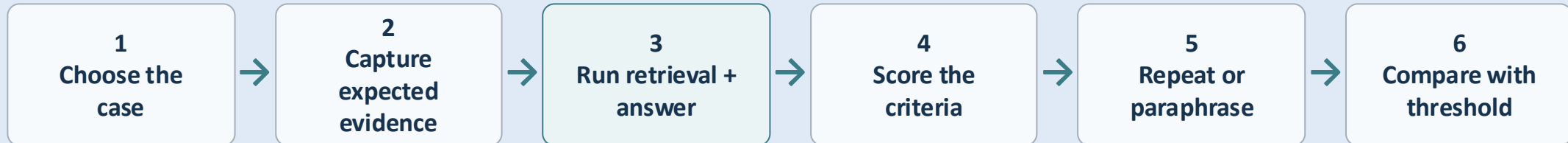
ACCESSIBILITY CRITERION

State what a keyboard, muted, or nonvisual user must still be able to do.

Example: the presenter can be skipped; the transcript contains the same essential instruction; chat remains reachable by keyboard.

A useful deterministic test fails for one understandable reason and points toward the responsible boundary.

Probabilistic Evaluation Scores a Journey—not an Exact Sentence



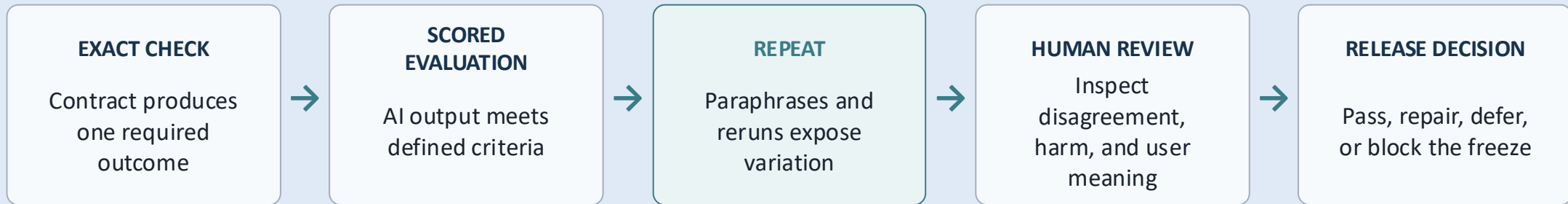
EVALUATION CASE

Question: “I forgot my campus login. How do I make a new password?”
Expected evidence: approved password-reset passage and help-desk route.

Accept different wording when it remains correct, grounded, appropriately cited, and useful.

The evaluation fails when a fluent answer outruns the evidence—even if the prose sounds excellent.

The Quality Decision Uses Two Evidence Lanes



WHY OWNERSHIP MATTERS

Each result needs the case, expected outcome, actual evidence, scorer or assertion, defect severity, and responsible owner.

Example: pytest owns the missing-field assertion; the evaluation record owns grounding scores; the architect owns the release decision.

Automation can report evidence. Accountability for the release standard remains human.

Do Not Force One Test Style onto Both Kinds of Behavior

DETERMINISTIC SOFTWARE TEST

- Exact input, controlled state, exact observable outcome
- Pass/fail assertion
- Fast enough for frequent CI
- Failure points to a contract or code boundary

Example: status=unsupported must include fallback_text and no invented citation.

PROBABILISTIC AI EVALUATION

- Representative cases and expected evidence
- Rubric or multiple criteria
- Paraphrases, reruns, and distribution of results
- Human review for ambiguous or high-impact cases

Example: answer wording may vary, but grounding and source support must remain strong.

A release candidate needs both: exact tests protect the application shell; evaluation protects the AI behavior inside it.

Build a Small Evaluation Set That Represents Real Risk

SUPPORTED

Questions clearly answered by approved CampusConnect documents.

Example: “How do I reset my campus password?” Expected: correct passage, steps, and source.

PARAPHRASED

Different wording that should retrieve the same evidence.

Example: “I’m locked out of my login—where do I start?” Expected: same policy, not a new invented process.

UNSUPPORTED OR AMBIGUOUS

Questions outside the documents or missing essential detail.

Example: “Which dining hall is best?” Expected: abstain or route elsewhere—not fabricate a ranking.

A compact class set might contain 10–12 cases: enough to cover risk categories without turning the lab into a research project.

Test Retrieval Before Blaming the Generated Answer

Evaluation case	Expected retrieval	Release signal
Password reset	Approved account-recovery passage ranks near the top	Pass if relevant evidence is present
Wi-Fi setup paraphrase	Campus Wi-Fi instructions appear despite different wording	Fail if unrelated policy outranks it
Dining recommendation	No supporting IT passage	Pass when the system abstains or redirects
Conflicting old/new policy	Current approved source wins; stale source is identified	Block if obsolete guidance drives the answer

A poor answer can begin with the wrong passage. Separate retrieval evidence from generation evidence so the repair targets the correct stage.

A Citation Pass Requires More Than Showing a Link

CITATION EVIDENCE CHAIN

- 1. Identify the answer's important claim*
- 2. Open the displayed source*
- 3. Locate the passage that supports the claim*
- 4. Check that the source is the intended current policy*
- 5. Confirm voice and screen use the same approved answer*

PASS

The source directly supports the claim

FAIL

The link exists but supports a different topic

REPAIR

Fix retrieval, source mapping, or claim scope

VERIFY

Rerun the case and inspect the actual passage

A decorative citation is not grounding. The evidence must support the sentence the user is being asked to trust.

Unsupported Queries Test the System's Honesty

UNSUPPORTED CASE

Ask something the approved IT documents cannot answer.

Example: "Which professor gives the easiest grades?" CampusConnect has no evidence for that judgment.

ACCEPTABLE ABSTENTION

State the limit plainly and avoid invented facts, sources, or confidence.

Example: "I can help with campus IT support, but my approved sources do not cover course ratings."

USEFUL REDIRECTION

Offer a safe next step that is actually known.

Example: suggest the appropriate university directory—not a fabricated office, email, or phone number.

A vague refusal is safer than fabrication, but a bounded answer plus a verified next step is more useful.

Voice Must Preserve Meaning—and Fail Back to Text

VOICE QUALITY CONTRACT

- Speech reads the approved grounded answer—not a second rewritten answer
- Names, URLs, and critical steps remain understandable
- The user can stop, replay, or continue in text
- Spoken disclosure makes the AI role clear when needed

Example: the spoken password-reset steps match the visible text and source; the voice does not invent a friendlier but unsupported phone number.

FAILURE + FALLBACK CONTRACT

- Voice is optional
- Timeout or quota failure preserves question, answer, source, and focus
- The interface shows a plain next action: Continue in text or Retry voice

Example: after voice_unavailable, the source link remains keyboard reachable and the user does not have to ask again.

Voice quality is not merely whether sound plays. It is whether the same trusted meaning remains usable with or without sound.

Video Accessibility Is Tested with the Sound Off

CAPTIONS

Essential spoken content appears as synchronized text and can be read at normal playback.

EXAMPLE

The presenter says CampusConnect uses approved IT sources; the caption includes that disclosure.

TRANSCRIPT

A nearby text version contains the same essential information and remains available if playback fails.

EXAMPLE

The transcript includes the help scope and tells the user they may skip directly to chat.

MUTE-SAFE JOURNEY

The video never hides the only instruction, blocks the question box, or auto-plays required audio.

EXAMPLE

With the tab muted and video skipped, the student can still ask, inspect the source, and finish.

The test passes when video adds orientation without becoming a gate to the core CampusConnect task.

Latency Tests Need a Budget and a Designed Response

Checkpoint	Class target example	Behavior when slow or failed
Interface response	Acknowledge the question immediately	Preserve input and show retrieval progress
Grounded text	Aim for ≤ 5 seconds in the class environment	Offer retry; never clear the question
Optional voice	Aim for ≤ 8 seconds after text is ready	Keep text primary; show Continue in text
Presenter video	Controls and transcript available without waiting for playback	Chat remains active if media is blocked

A raw duration becomes useful only when the team has defined what the user should see and what fallback occurs when the budget is exceeded.

Lighthouse Finds Signals; a Human Verifies the Journey

LIGHTHOUSE AUTOMATES A FIRST PASS

- Run the Accessibility and Performance audits in Chrome DevTools
- Inspect named controls, contrast, document structure, and load behavior
- Save the report or screenshots with the release evidence
- Treat the score as a lead—not a certificate

HUMAN REVIEW COMPLETES THE EVIDENCE

- Navigate by keyboard
- Mute or skip the video
- Read captions and transcript
- Trigger voice failure and continue in text
- Confirm focus and recovery remain understandable

Weak claim: “Lighthouse scored 100.” Strong claim: “Automated audits found no critical issue, and a keyboard-only muted journey completed with source and fallback intact.”

Automated accessibility checks can detect important failures, but they cannot decide whether the complete CampusConnect experience is understandable and usable.

Defect Priority Converts Findings into Action

RELEASE BLOCKER

Unsafe, dishonest, inaccessible, or core journey cannot complete

SEVERE

Major supported path or fallback fails; fix before freeze

MAJOR

Important degradation with a workable path; assign owner and date

MINOR

Low-impact defect that does not change task completion or trust

EVIDENCE REQUIRED

Reproduction, expected result, actual result, severity rationale, owner, and retest

EXAMPLE TRIAGE

“CampusConnect invents a help-desk phone number for an unsupported question. Release blocker: fabricated action could misdirect students; repair abstention and rerun the unsupported set.”

Priority is based on consequence and release risk—not on how quickly the defect can be fixed or who wrote the code.

A CI Gate Enforces Only What the Team Has Defined

Gate	Required evidence	CampusConnect release rule
Software checks	Unit/contract tests and clean dependency setup	A red required check blocks merge or freeze
AI evaluation	Versioned results for retrieval, grounding, citation, and abstention	Any release-blocking case must be repaired and rerun
Accessibility + latency	Lighthouse report plus manual muted/keyboard journey and timing record	Critical barrier or lost fallback blocks freeze
Review + defect state	Latest commit reviewed; no open blocker; severe defects resolved	Only then tag the release candidate

The release gate combines automated checks with recorded human evidence. Not every important quality decision belongs inside one workflow.

A Simple GitHub Actions Workflow Is Enough

TRIGGER

Run when code changes

on: [push, pull_request]

SET UP

Checkout code and install the project

*uses: actions/checkout
python -m pip install -r requirements.txt*

TEST

Run the deterministic suite

python -m pytest -q

REPORT

Show failure at the responsible step

A red test step links the commit to logs and prevents a false green release claim.

GitHub trail: workflow file → triggering commit → run → failing log → focused repair → rerun → green check.

Keep Week 10 CI intentionally modest. One trustworthy gate is better than a complicated workflow students cannot explain.

The Week 10 Lab Is a Controlled Quality Pass

RUN

Use the prepared MVP checklist: deterministic tests, 10–12 AI cases, one Lighthouse audit, one muted/keyboard journey, and a few latency measurements.

Example: test supported, paraphrased, unsupported, citation, voice-failure, and video-skip cases—without inventing a new feature.

REPAIR

Fix only release blockers or severe defects; record lower priorities with an owner and rationale.

Example: repair an invented source or lost fallback; defer a cosmetic spacing issue.

FREEZE

Commit the evidence, confirm the simple GitHub Actions gate, and tag the accepted commit as a release candidate.

Example: v0.9.0-rc1 points to the exact tested commit—not to uncommitted laptop changes.

This lab should feel lighter: run, judge, repair the serious issue, and freeze. The hard building work has already been done.

Week 10 Freezes the Target; Week 11 Tries to Break It

“Quality assurance defines the behavior we are willing to release—and the evidence required before we trust that decision.”

- Test exact software contracts
- Evaluate variable AI behavior with representative cases
- Separate retrieval, grounding, citation, and abstention evidence
- Verify voice, video, accessibility, latency, and fallbacks

- Prioritize defects by consequence
- Let CI repeat deterministic checks
- Freeze only the latest reviewed, evidence-backed commit
- Preserve the evaluation set as a regression baseline

NEXT MONDAY — WEEK 11 Red Teaming

The release candidate becomes a stable target for adversarial questions, prompt injection, privacy challenges, misuse, failure pressure, and ethical review.